NETWORKING



# Scaling AI Beyond the Cloud Boundary: High-Capacity Interconnects Between Hyperscalers and GPUaaS NeoClouds




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# AI scale crosses the cloud boundary

Demand outruns power, space, and GPU lead time — so capacity is rented, not built. Data gravity stays put.

HYPERSCALER SERVICES

## Services keep scaling

Storage, control, and reachability stay inside.

- **STORAGE** — datasets + checkpoints
- **VPN / ON-PREM** — private reachability
- **PLATFORM SERVICES** — control plane + telemetry

DATA STAYS HERE

AI WORKLOAD GROWTH

**Two operators. One path.**

**Trust + Control boundary**

This is not just DCI for packets.
It is DCI for service reachability.

RENTED GPU CAPACITY

## Compute moves outside

- **GPUaaS** — external operator
- **NeoClouds** — leased accelerators

**ARCHITECTURE QUESTION ?**

**How do independently operated domains share cloud services at AI scale without exposing internal infrastructure?**

Translate service intent, not just packets.

# Requirements for a service-aware interconnect edge

**WHY EXISTING DCI IS NOT ENOUGH**

Traditional DCI moves traffic across a boundary. AI-era DCI must preserve cloud service behavior and expose reusable OCP platform requirements.

**Service-aware edge requirements**



**Multi-tenant isolation at the edge**

**Tunnel and route scale**

**High-density ECMP + consistent hashing**

**Programmable packet transformation**

**Control-plane scale**

**Deterministic recovery + deep buffers**

# Cisco 8223 with Silicon One P200: a deep-buffered, secure programmable edge for this role

## One ASIC. One Pipeline. One Power envelope.




**51.2 Tbps-class fixed system**
**Deep-buffered WAN, DCI/Scale Across, and AI DC edge**

| 64x800G | 3RU | ZR-optics | MACsec |
|---|---|---|---|
| dense ports | fixed system | Long Haul DCI | Secure |

### POWER EFFICIENCY

Most power-efficient 51.2T deep-buffer router, keeping scale-across growth inside a tighter power envelope.

### DEEP BUFFERS

Absorb bursts across two independently operated QOS domains.

### P4 PROGRAMMABILITY

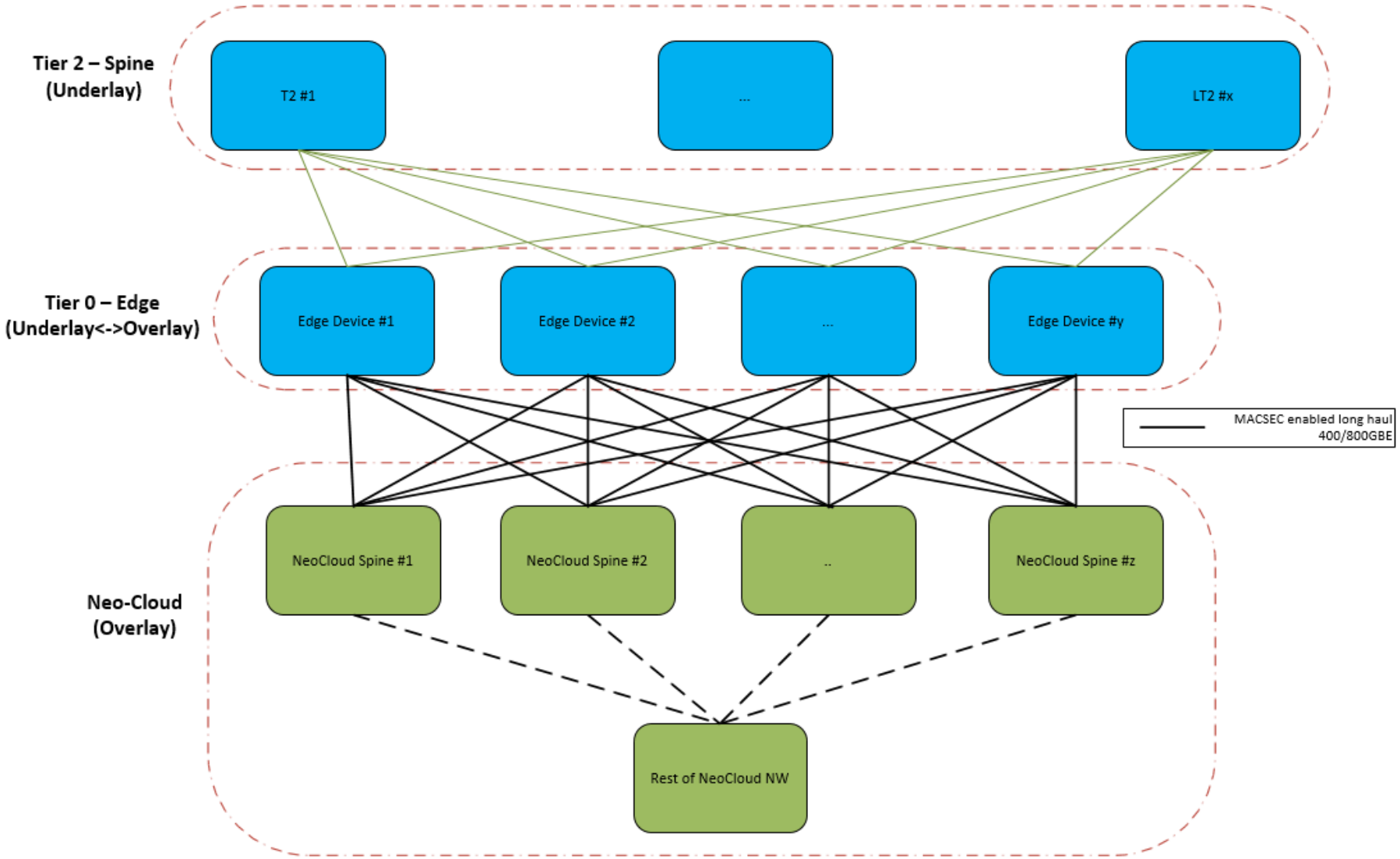
Post-ideation feature development and future proofing

## Service-Aware Edge

### What P200 brings to each requirement ?

| The requirement | What the platform brings |
|---|---|
| **Multi-tenant isolation at the edge** | SONiC VNET / VRF / VXLAN with per-tenant sub-interfaces, overlapping subnets, and SDN-driven provisioning. |
| **Tunnel + route scale** | 256K VXLAN tunnel next-hops on the delivered role, with headroom toward 1M-class edge scale. |
| **High-density ECMP + consistent hashing** | 2048-way tunnel ECMP with per-next-hop rewrite, plus stable hashing buckets through member change. |
| **Programmable packet transformation** | P4 pipeline for inner-DST-MAC + VNI rewrite, VXLAN port control, encap, and address-family change. |
| **Control-plane scale** | Per-VNET dynamic BGP at 2K+ sessions, with BGP and link-state tracking feeding the SDN layer. |
| **Deterministic recovery + deep buffers** | 51.2 Tbps-class fixed system with deep buffers, MACsec, config replay, and fast route programming. |

# Physical Network Architecture



# Physical Network – Unique Requirements

## WHAT THIS LAYER MUST SOLVE

**Long-haul reach, bursty AI traffic, and multiple-edge devices used in an Active-Active fashion result in unique requirements on the physical network**

### Deep Buffer

Handle bursty training traffic

Handle lots of in-flight data over long-haul links

### Macsec

Secured long haul traffic

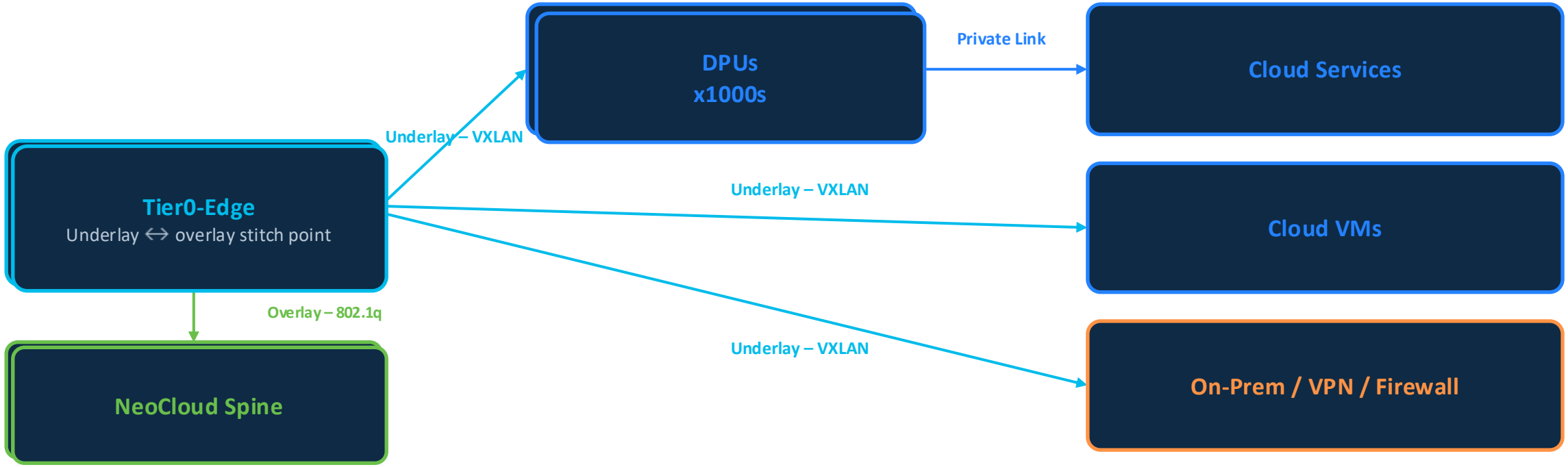
### Reliable Config Replay

Reliably replay config at high config scale, avoiding premature datapath enablement upon image upgrade, device restart etc.

### HA and Failure handling

Propagate failures across 2 dis-joint networks( underlay and overlay) by using;
Underlay BGP session tracking
Overlay Link state tracking

# Logical Network Architecture



# Logical Network – Unique Requirements

## OVERLAY / UNDERLAY REACHABILITY

**Tier0-Edge is a meet-me point for underlay <-> overlay termination at scale**

### Programmability for various parts of the VxLan header at scale

Inner SMAC, DMAC and outer VNI rewrite per route at >250k scale to support a massive number of endpoints which need to be reachable to/from Neocloud GPUs

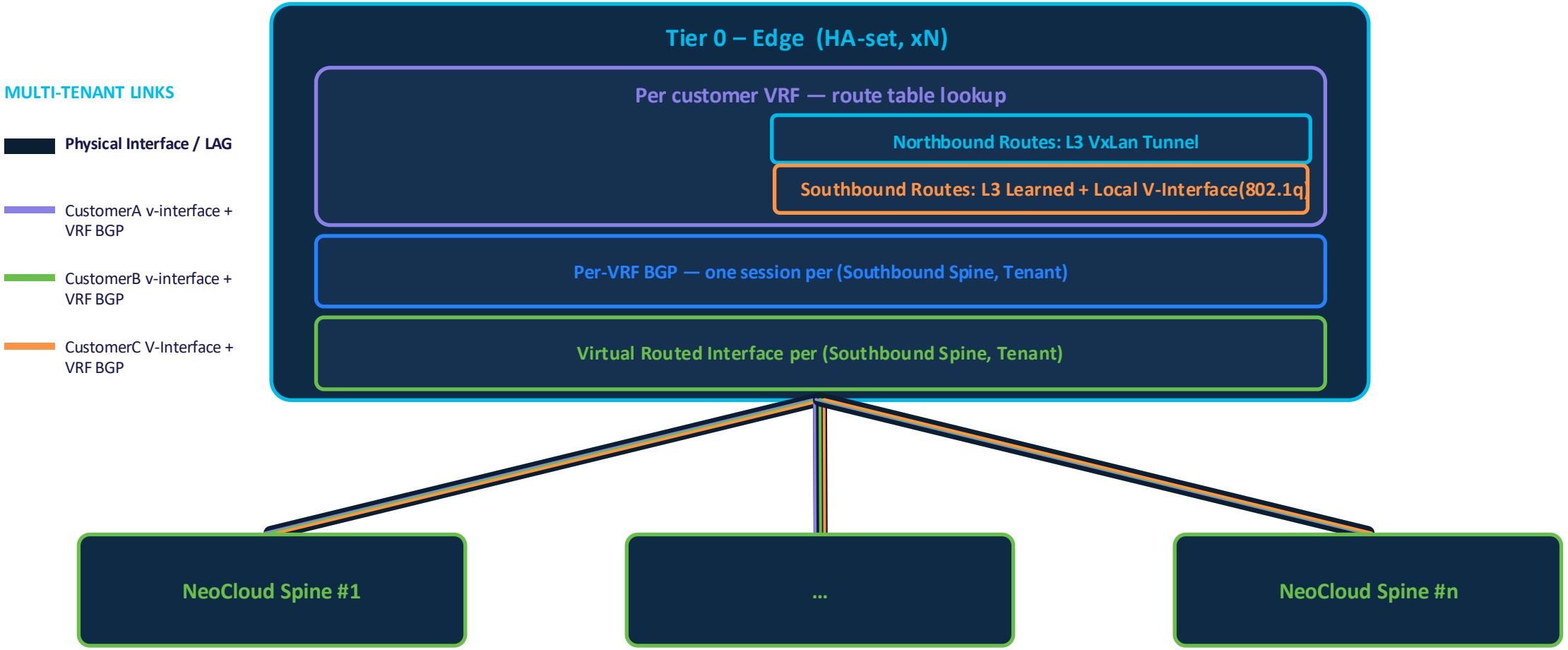
### High-Density Overlay ECMP

Distribute AI traffic across large overlay->underlay path counts (2k+)

### Consistent Hashing

Reduce stateful flow impact upon DPU failure

# Device Networking View



# Device Networking – Unique Requirements

### BGP session scale

High scale of per-vrf BGP sessions(1000s)

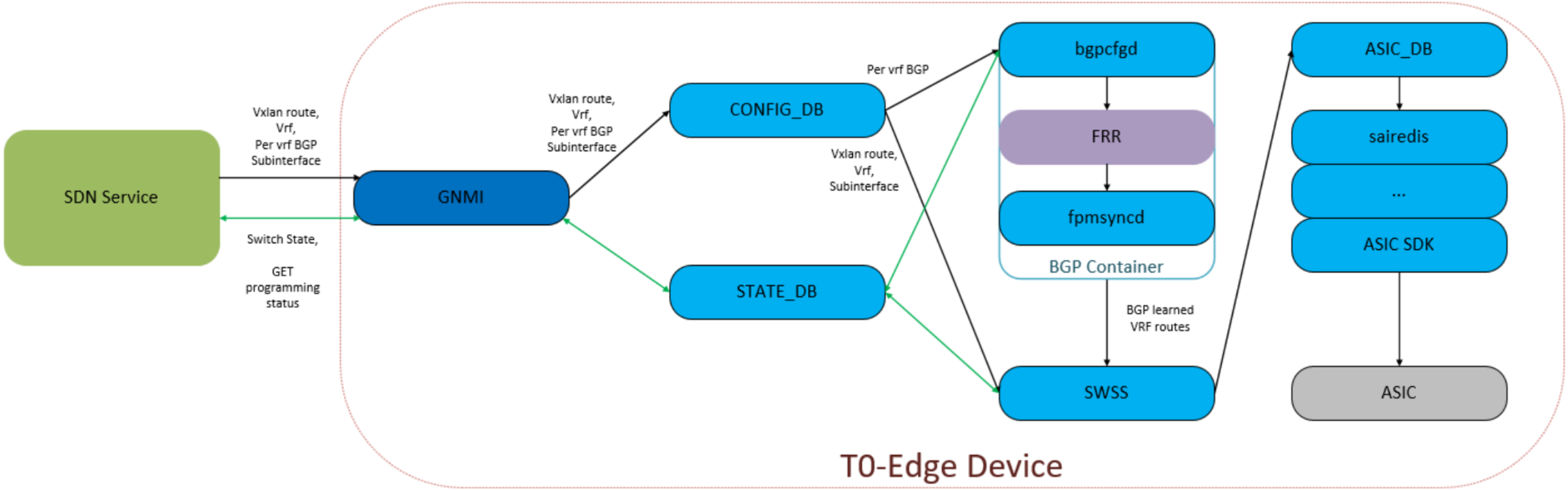
### Multi-tenancy

Isolate VNETs, VRFs, overlays, and tenant policies.

### Route scale

Support large tenant and service route tables across a lot of VRFs

# SONiC Software Architecture and SDN programming



# This use case maps to OCP workstreams

**OCP RELEVANCE**

The session is relevant to OCP because this use case stresses exactly the hardware, SAI, SONiC, and routing behaviors needed for AI-era interconnects.

| | |
|---|---|
| **OCP hardware** | Deep-buffered edge platforms, dense 800G interfaces, route scale, tunnel scale, ECMP scale, and telemetry. |
| **OCP SAI** | Abstractions for VXLAN, VRF, ECMP hashing, route programming, and packet rewrite behavior. |
| **SONiC routing** | Faster config application, route programming, config replay after restart, and 1.5K+ BGP sessions. |
| **OEM / ODM adoption** | A repeatable gateway capability set that can be implemented across deep-buffered platforms. |

**Community ask: collaborate on routing scale, resiliency, SAI behavior, and deep-buffered gateway requirements.**

# Call to Action

OCP COMMUNITY ASK

**Help make AI-era cloud-to-NeoCloud interconnects broadly adoptable across open platforms.**

### Faster route programming

Reduce time to push routes and tenant configuration into the data path.

### Reliable replay and recovery

Improve BGP and config replay behavior after process or system restart.

### BGP scale beyond 1.5K

Support dense multi-tenant edge deployments without fragile control-plane behavior.

### SAI / SONiC packet behavior

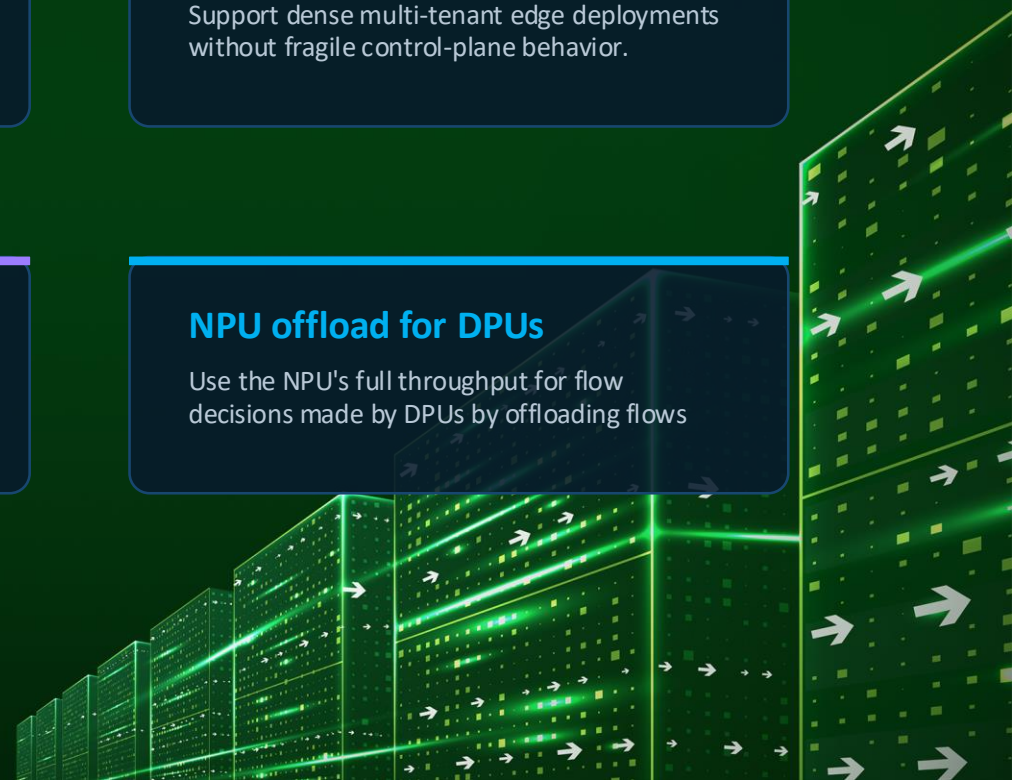
Strengthen abstractions for VXLAN, VRF, ECMP hashing, and packet rewrite.

### OEM / ODM adoption

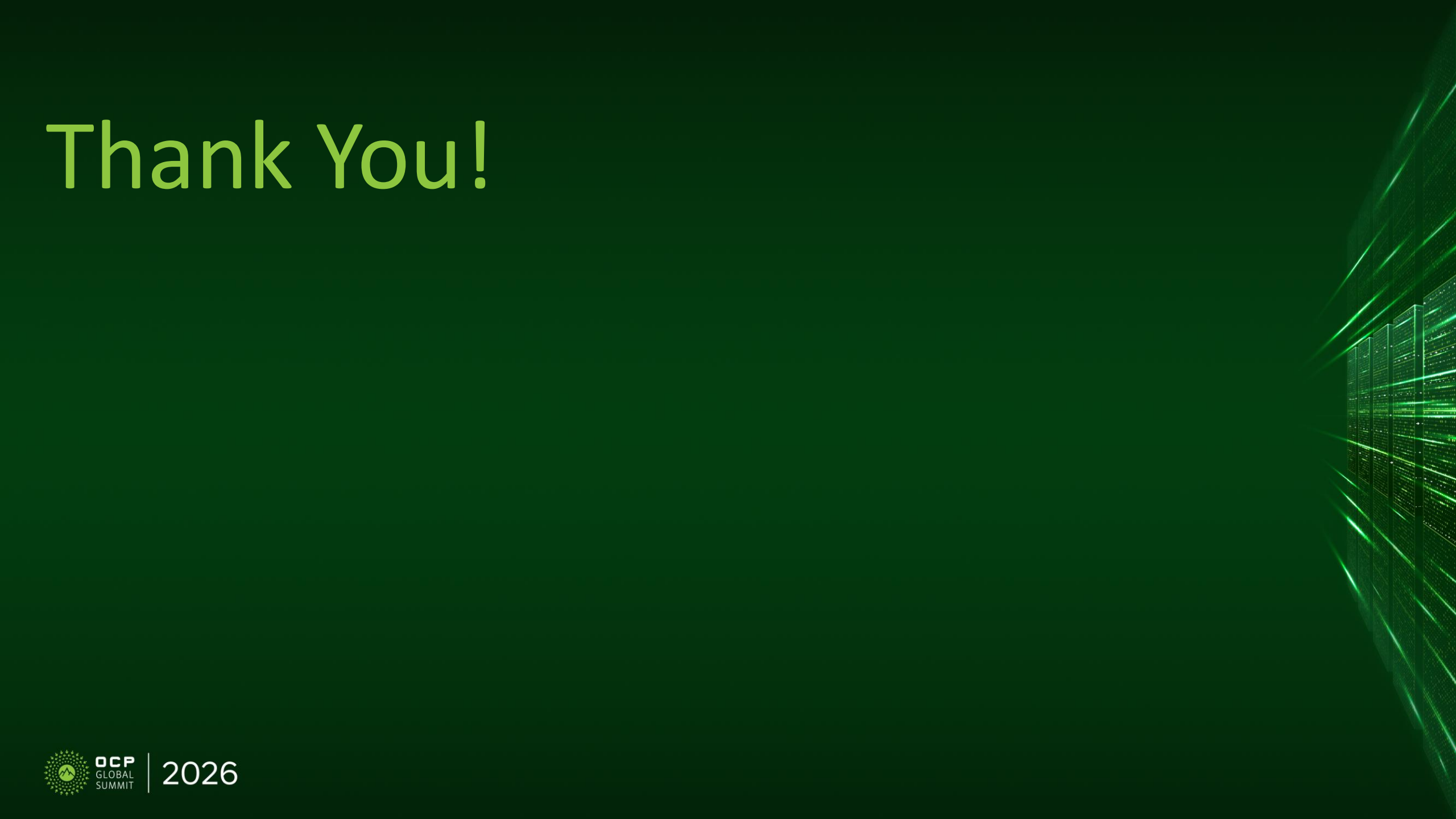
Make deep-buffered T0 edge gateway capabilities repeatable across platforms.

### NPU offload for DPUs

Use the NPU's full throughput for flow decisions made by DPUs by offloading flows



# Thank You!



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